

Adv Graphics Final Report

Depth of Field Rendering with Focus Control

Camille Jouanet

Real-Time Lens Blur Effects and Focus Control

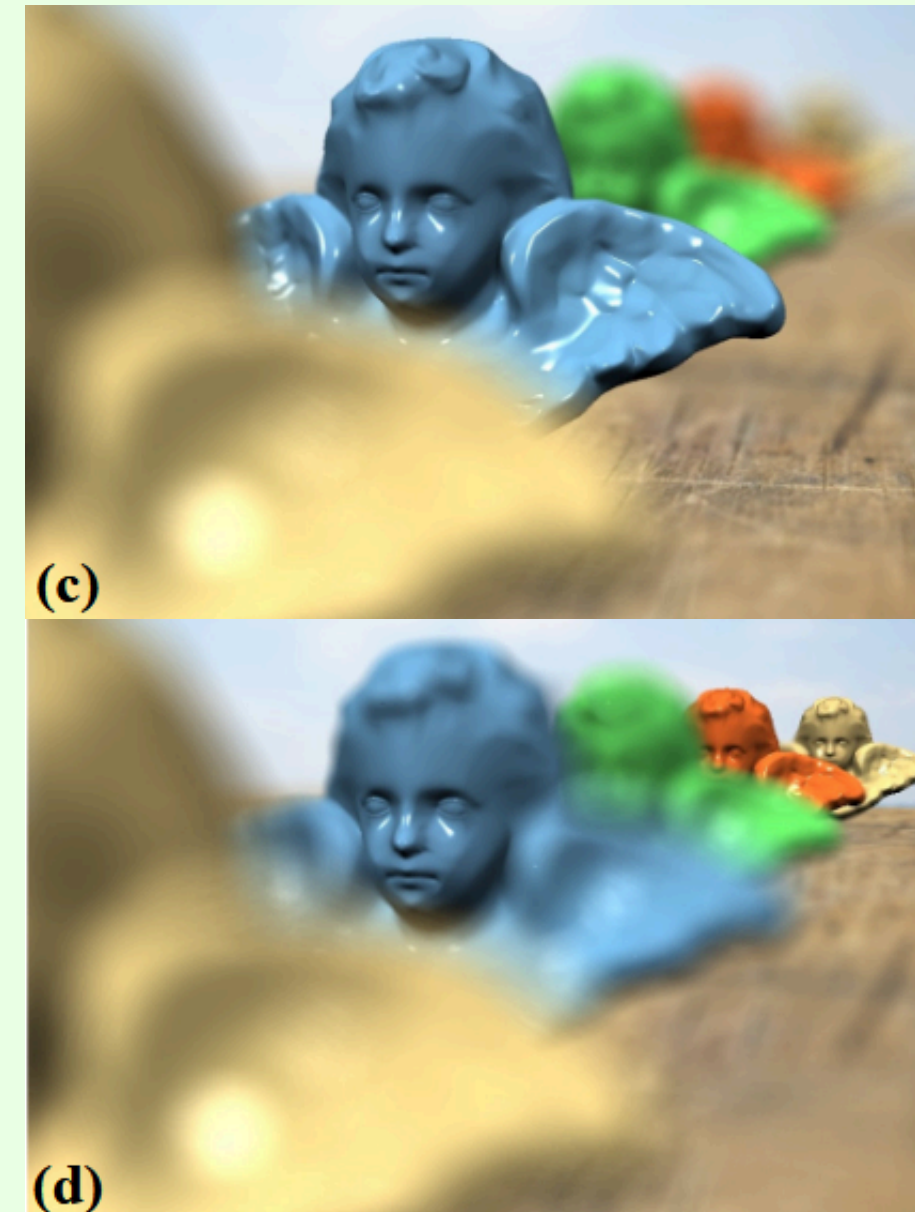
Sungkil Lee, Elmar Eisemann, Hans-Peter Seidel

A real-time accurate lens simulation:

- Focus control
- Bokeh (artistic blur)
- Optical aberrations

Aim to improve classical naive methods with
a powerful GPU ray tracing pipeline

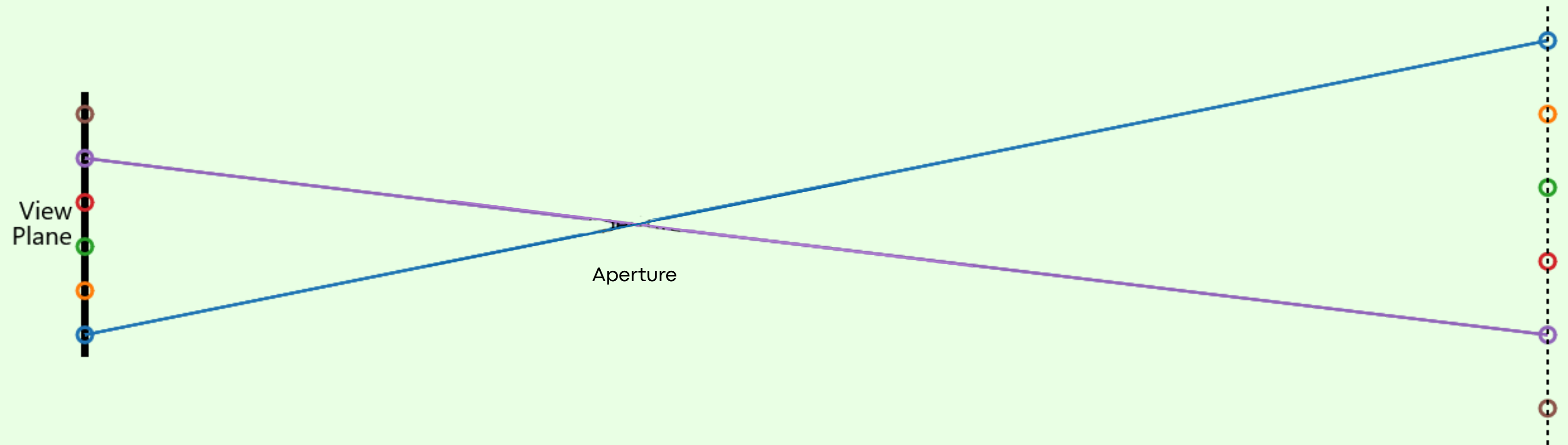
→ high cost for large CoC, intensity leak,
incorrect halo...



Lee et al. 2010

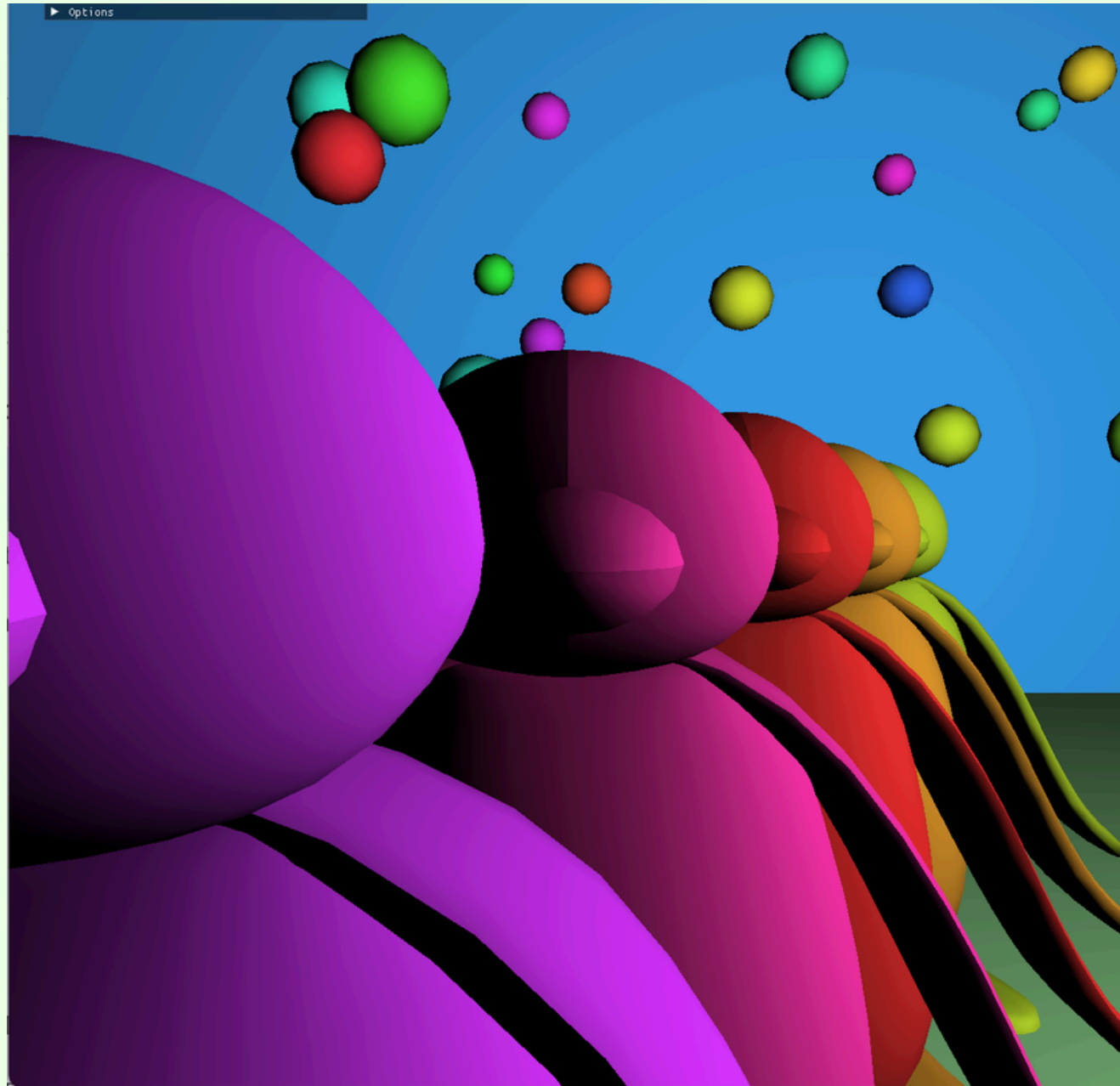
Lens Modelization

Thin lens

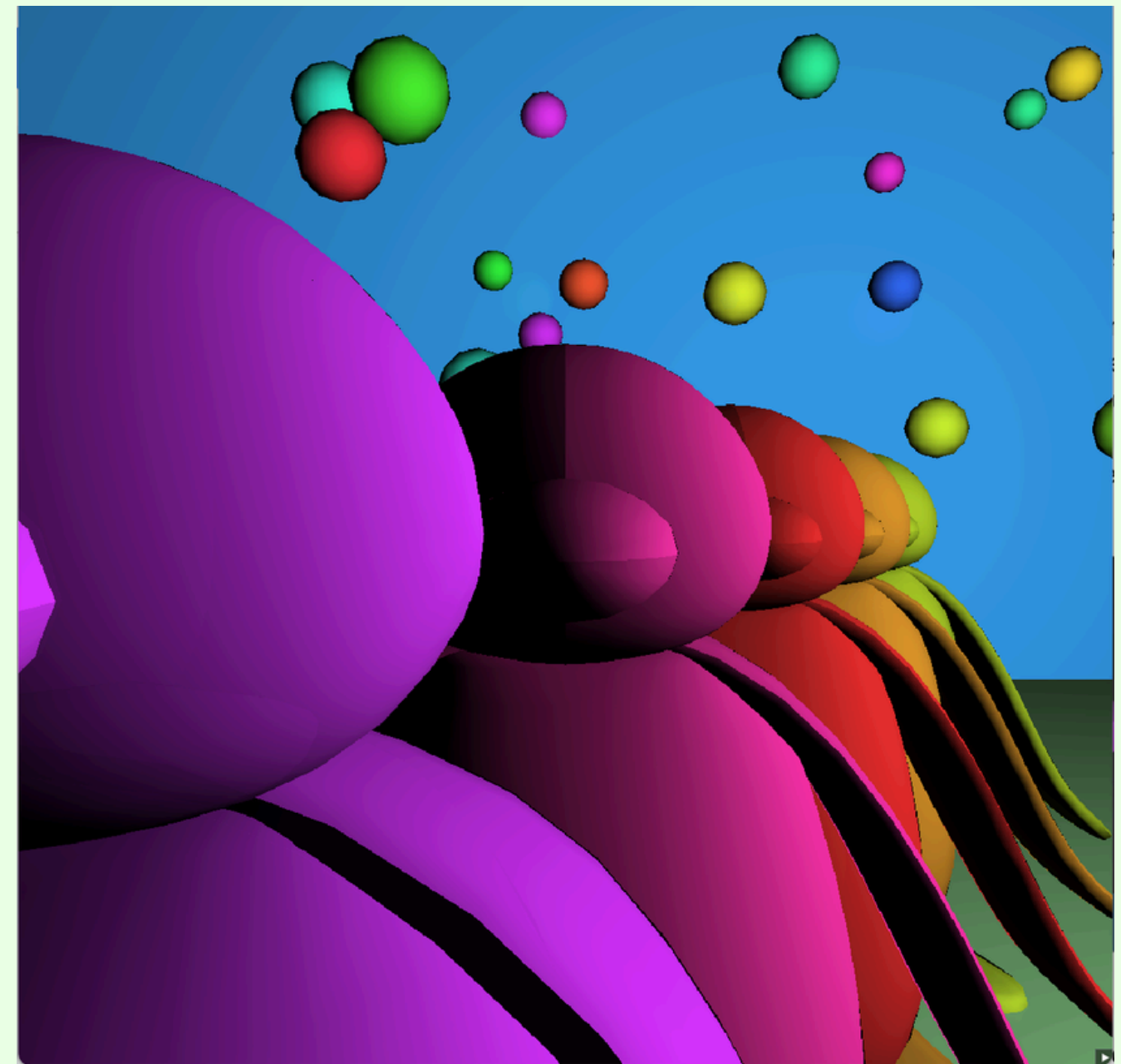


Pinhole image

Pinhole image



Original

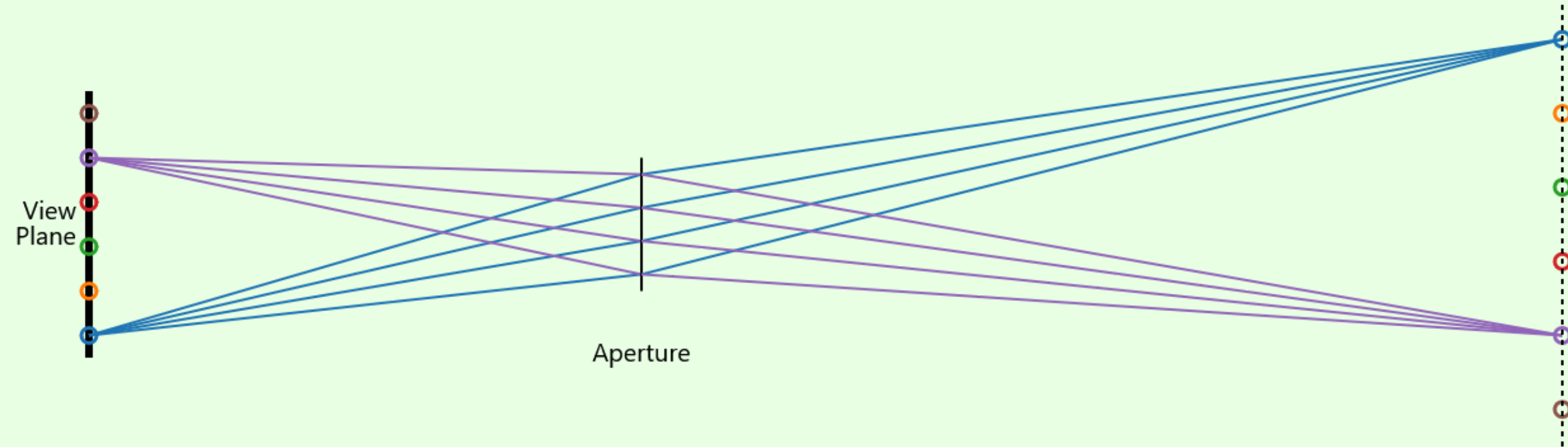


Pinhole

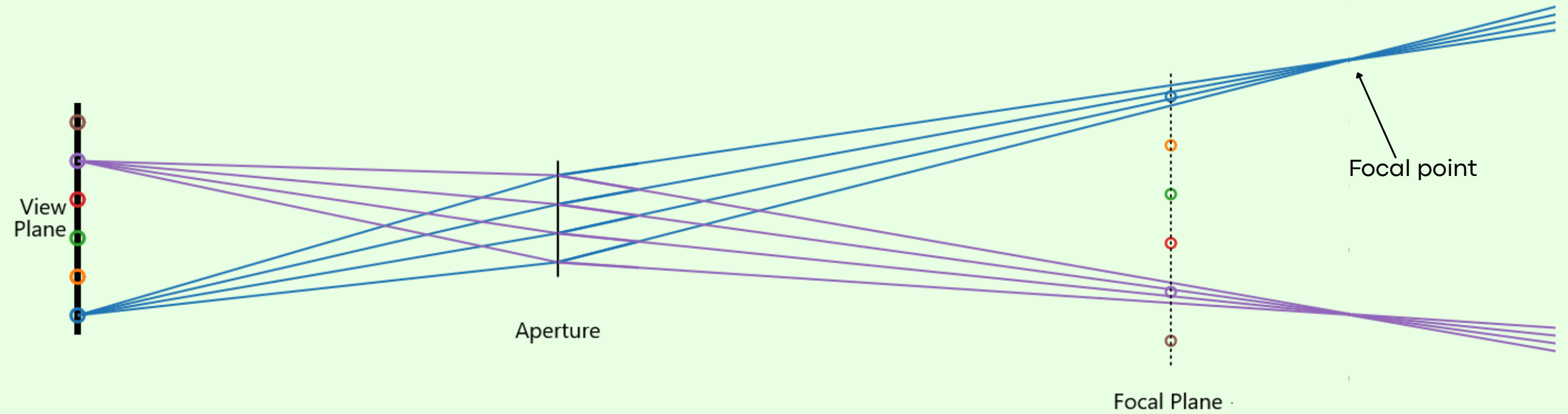
Lens Modelization

Thin lens

In focus



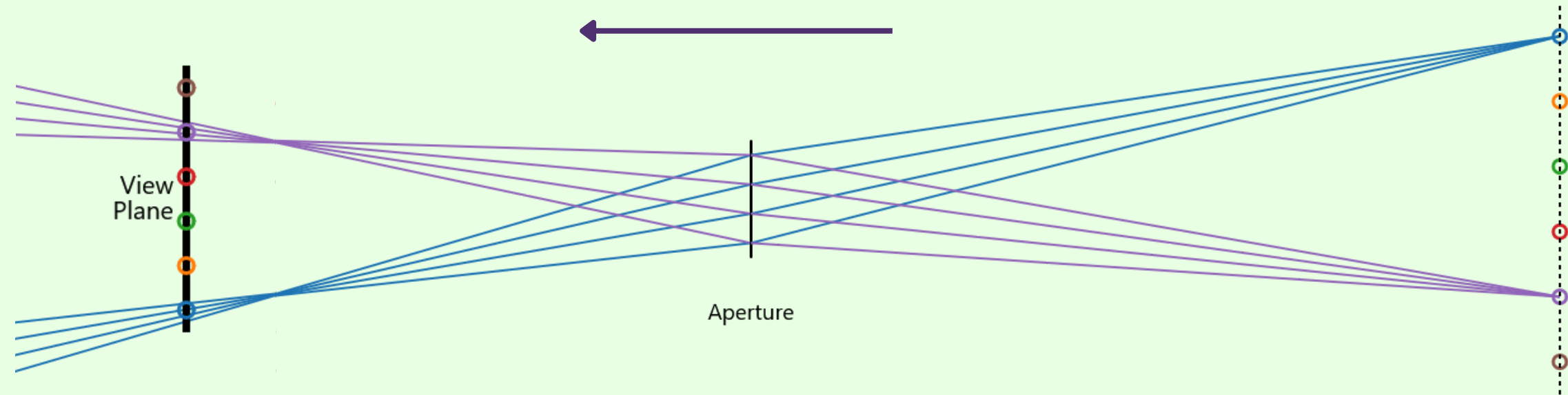
Out of
focus



Lens Modelization

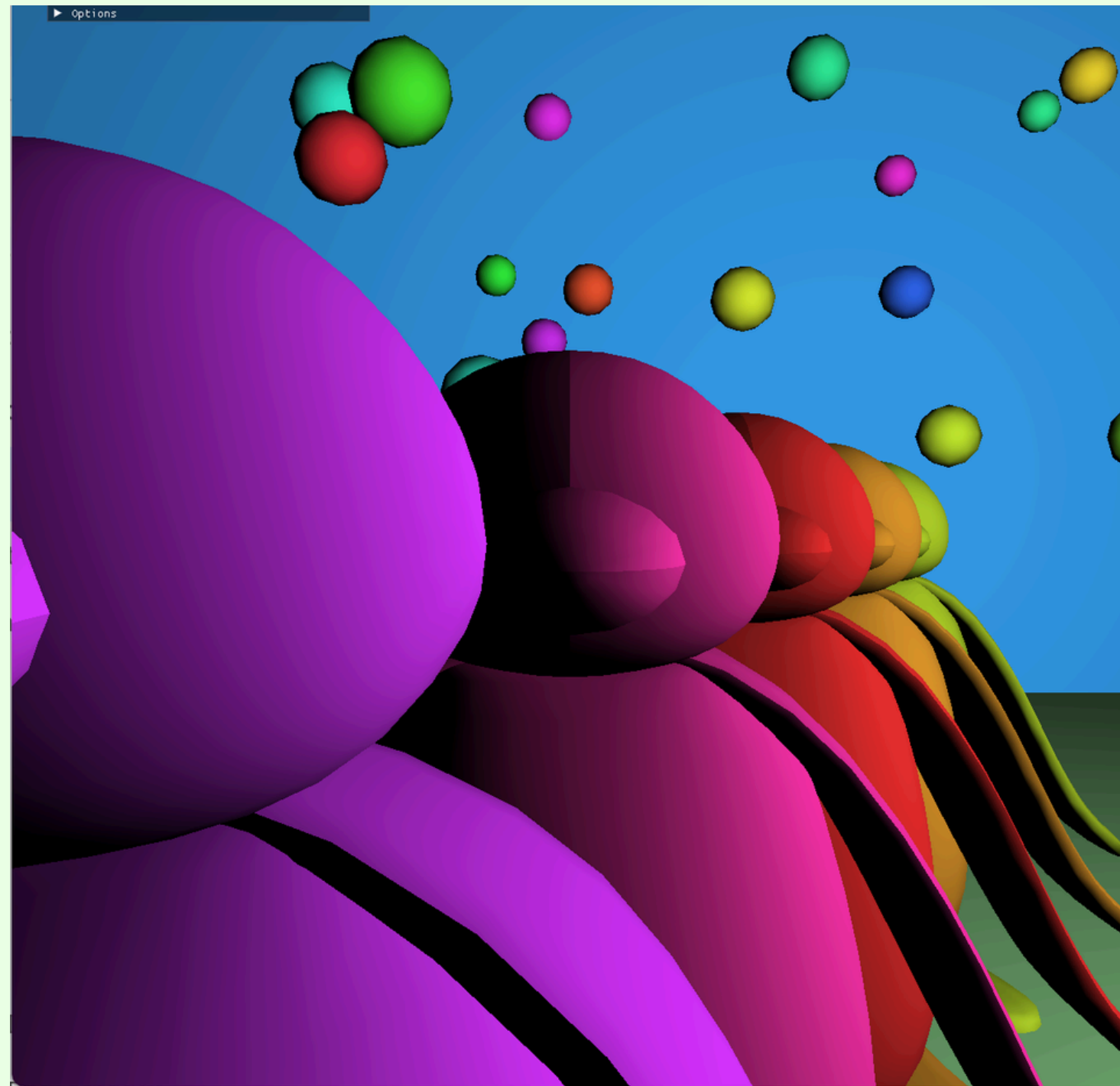
Thin lens

Our
model

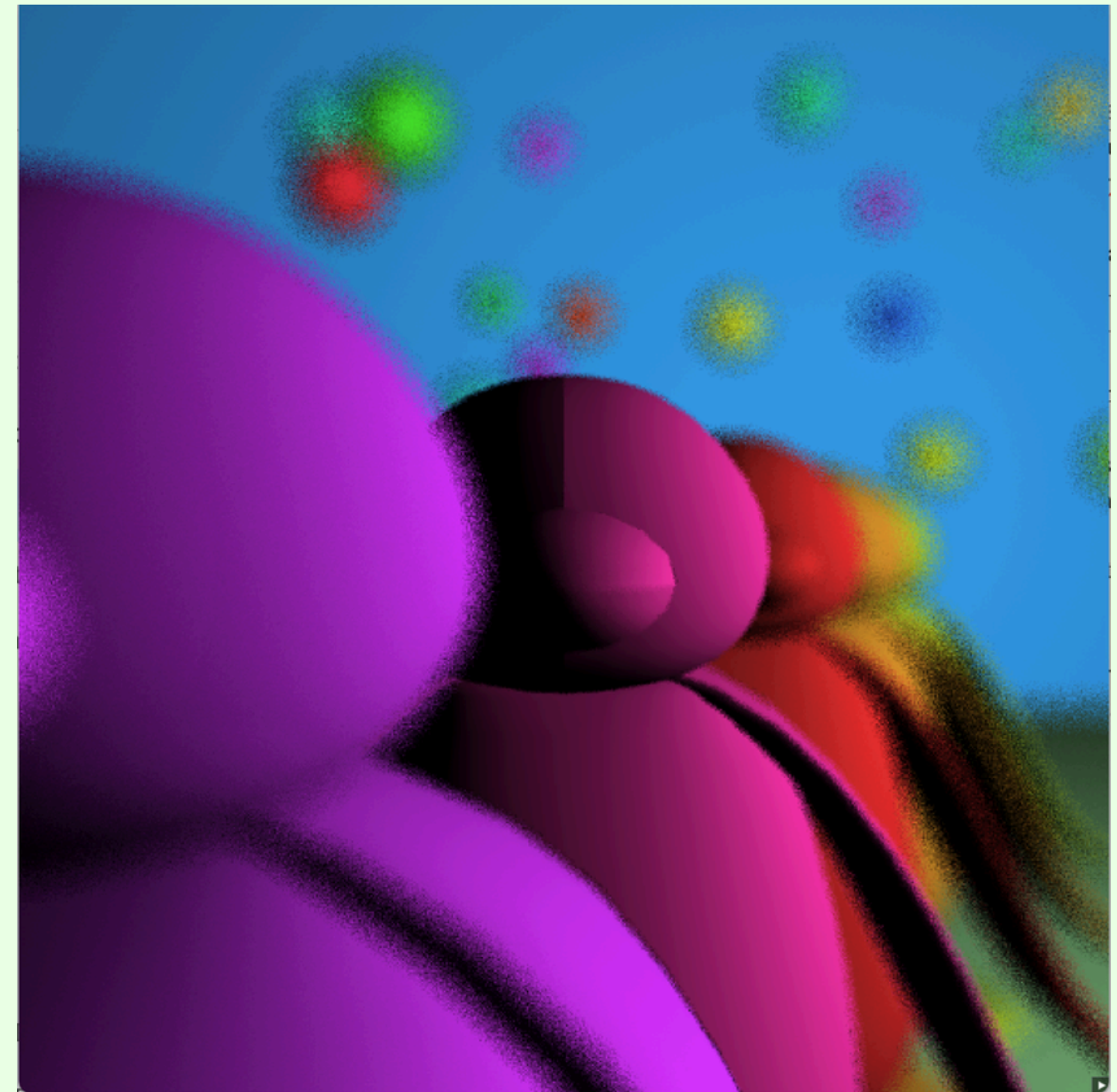


Depth of Field rendering

N=16, uniform distribution



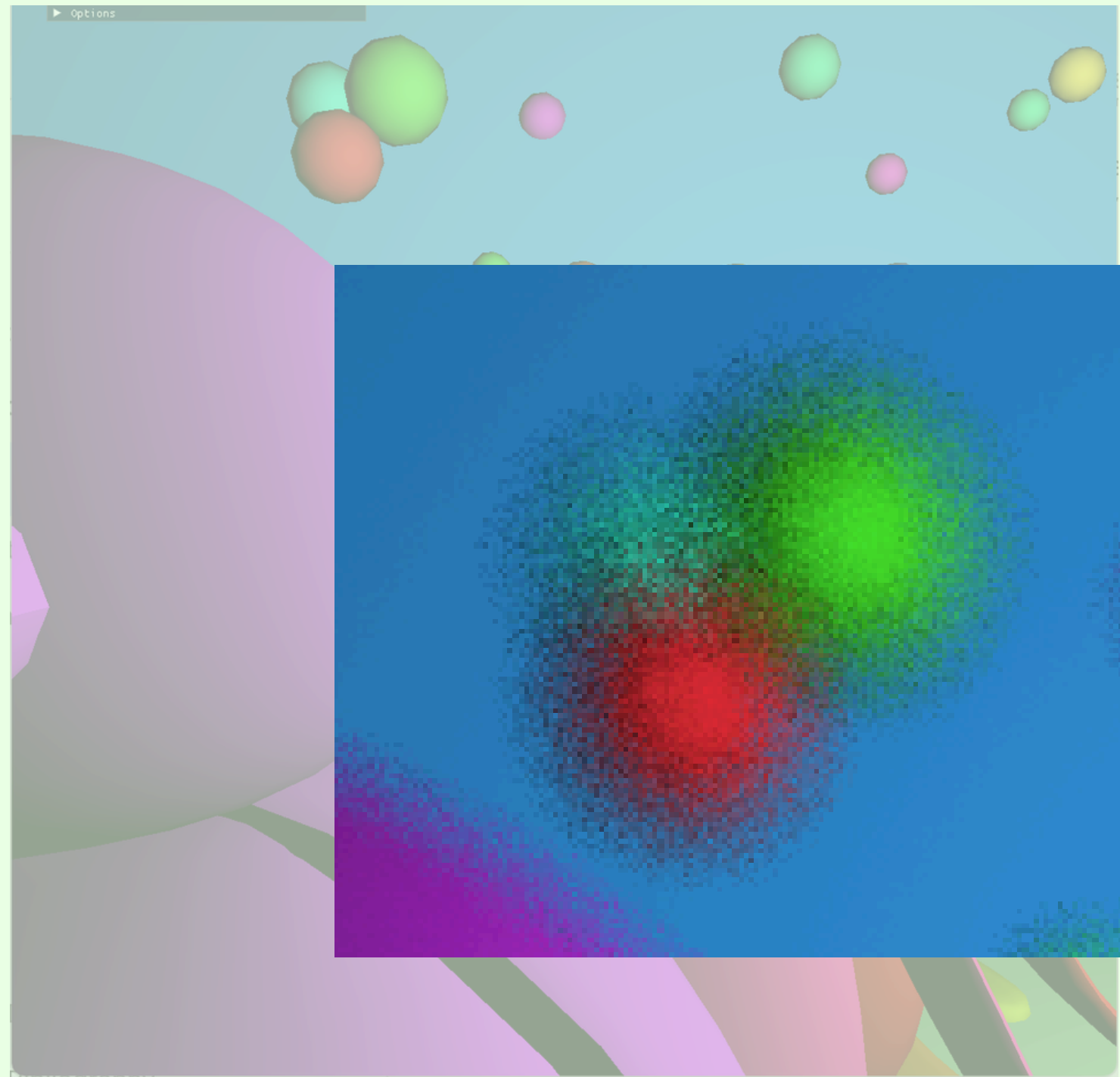
Original



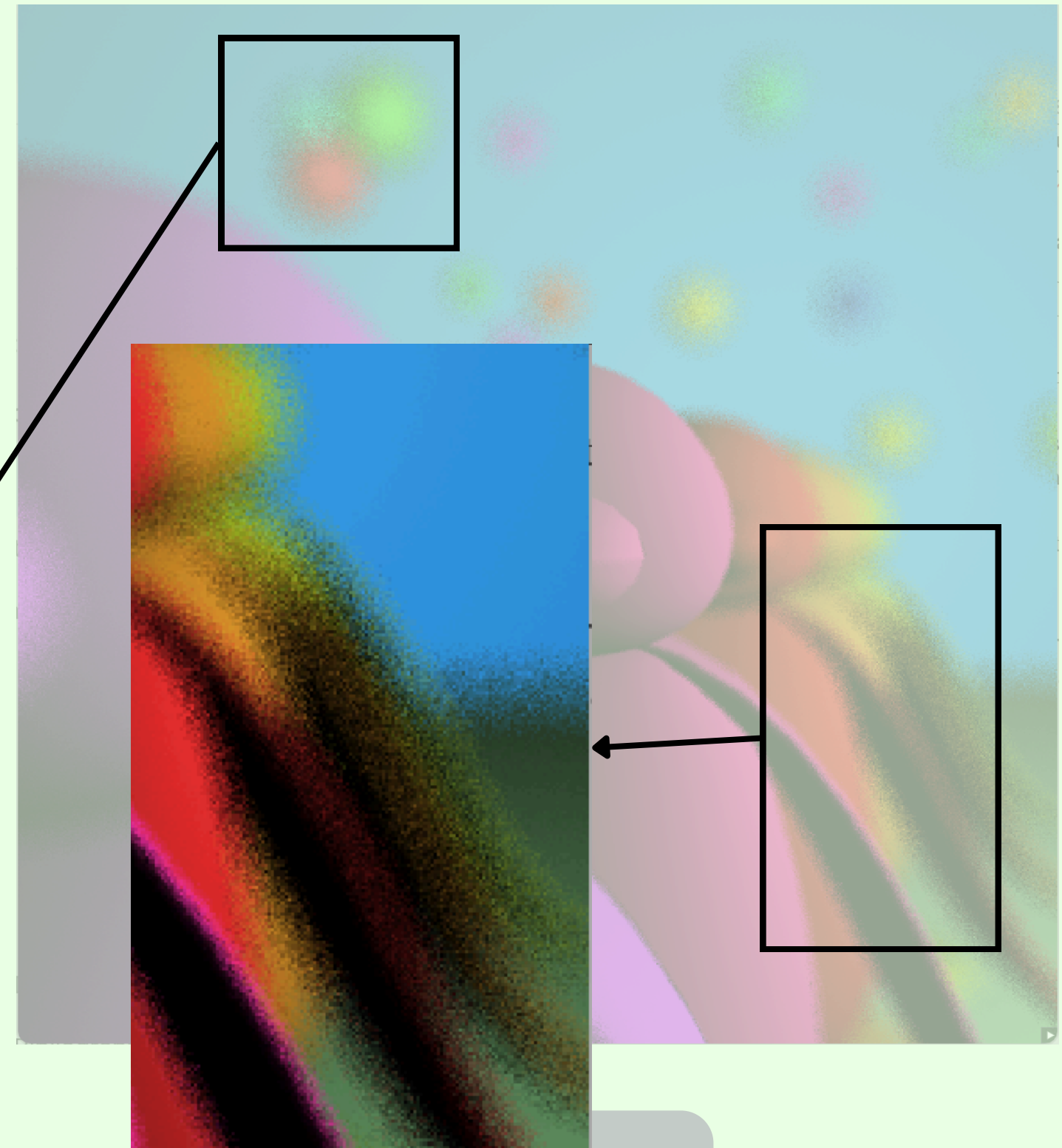
DoF render

Depth of Field rendering

N=16, uniform distribution



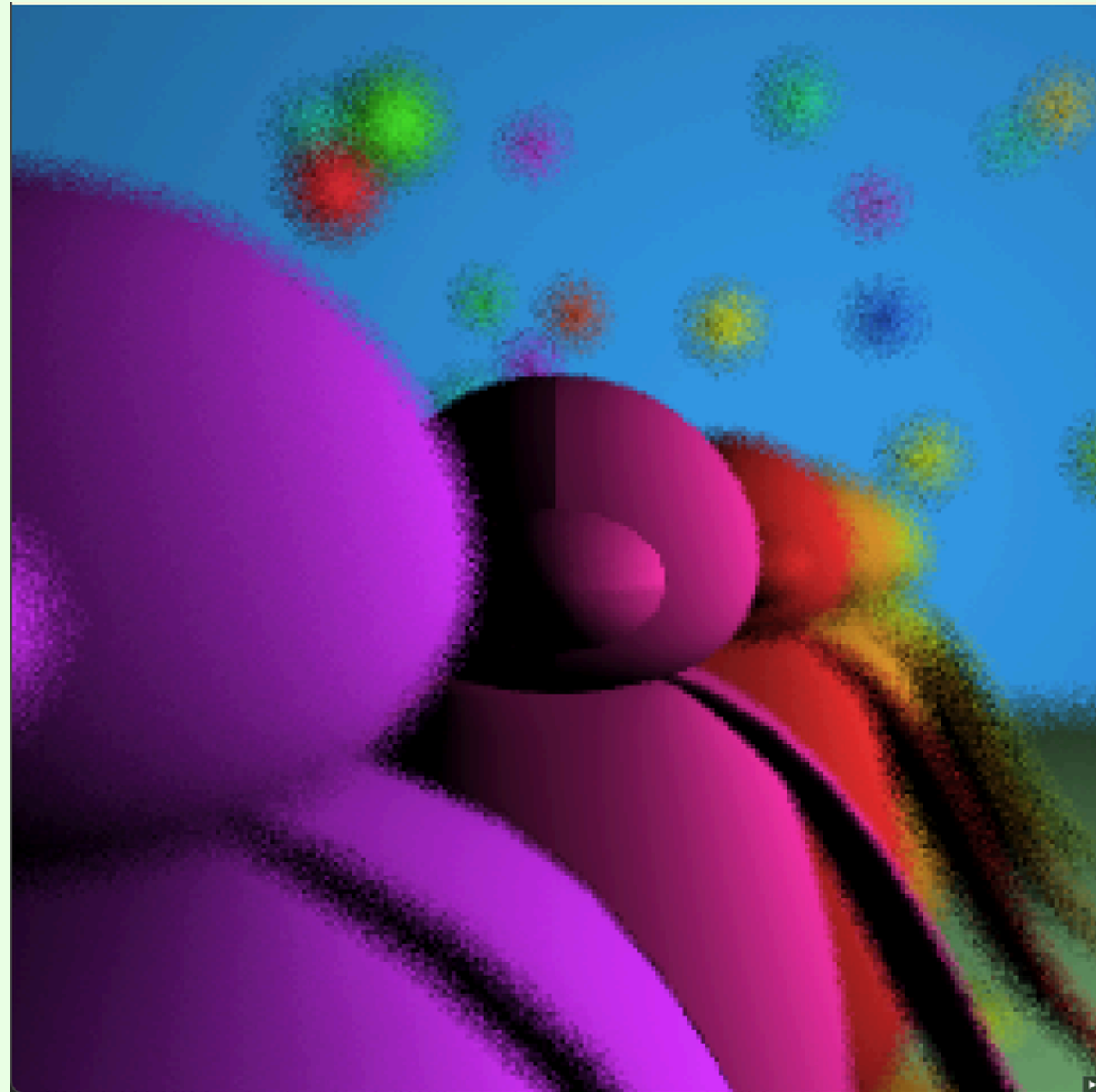
Original



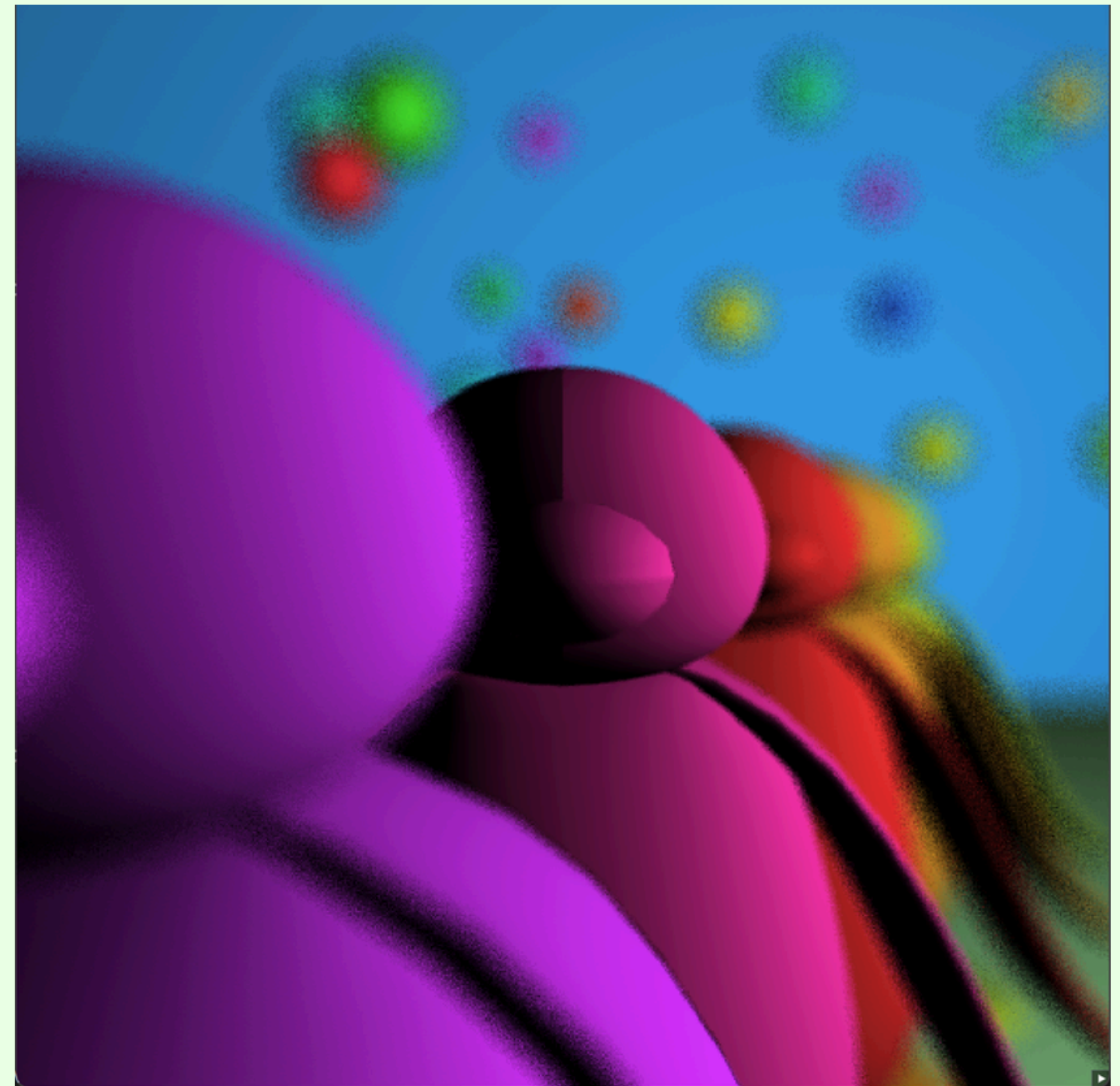
DoF render

Sampling Methods

N=16, uniform distribution



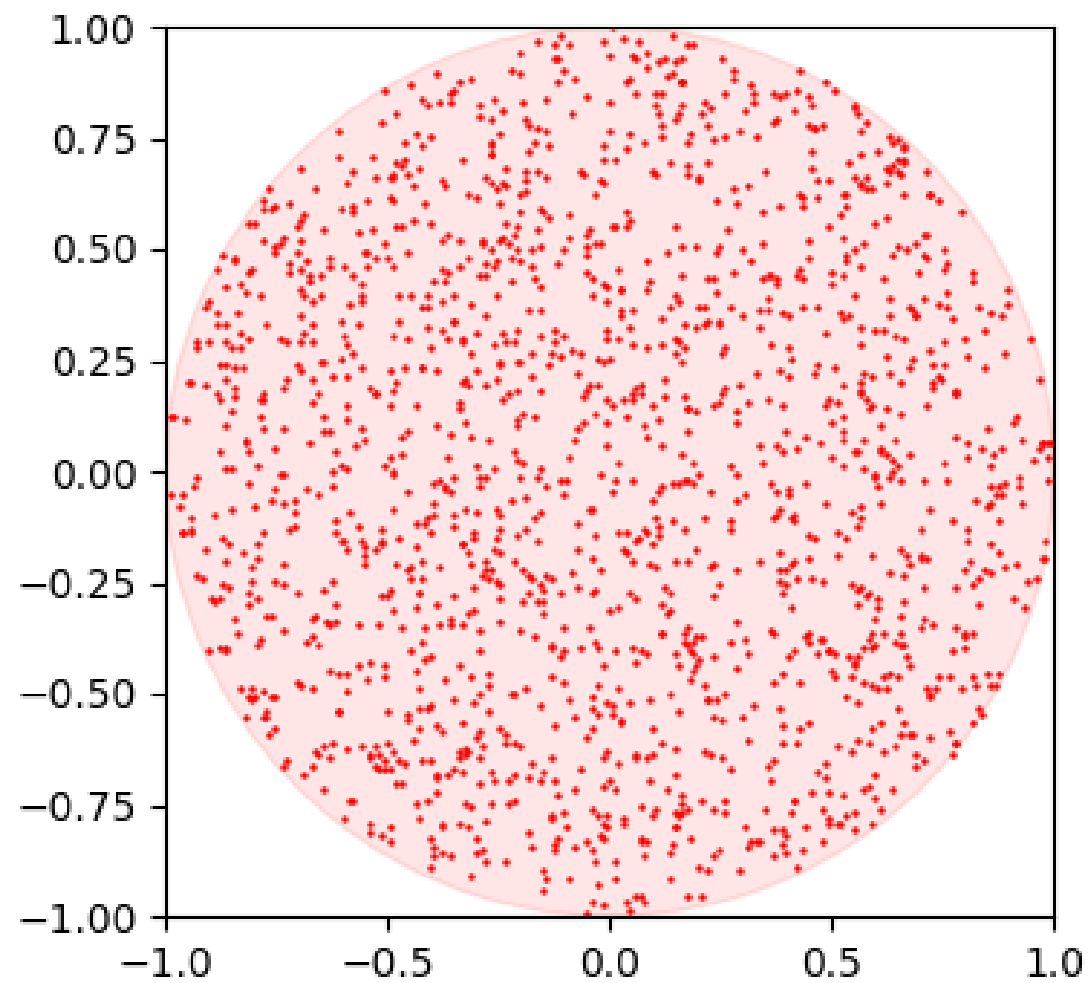
Random



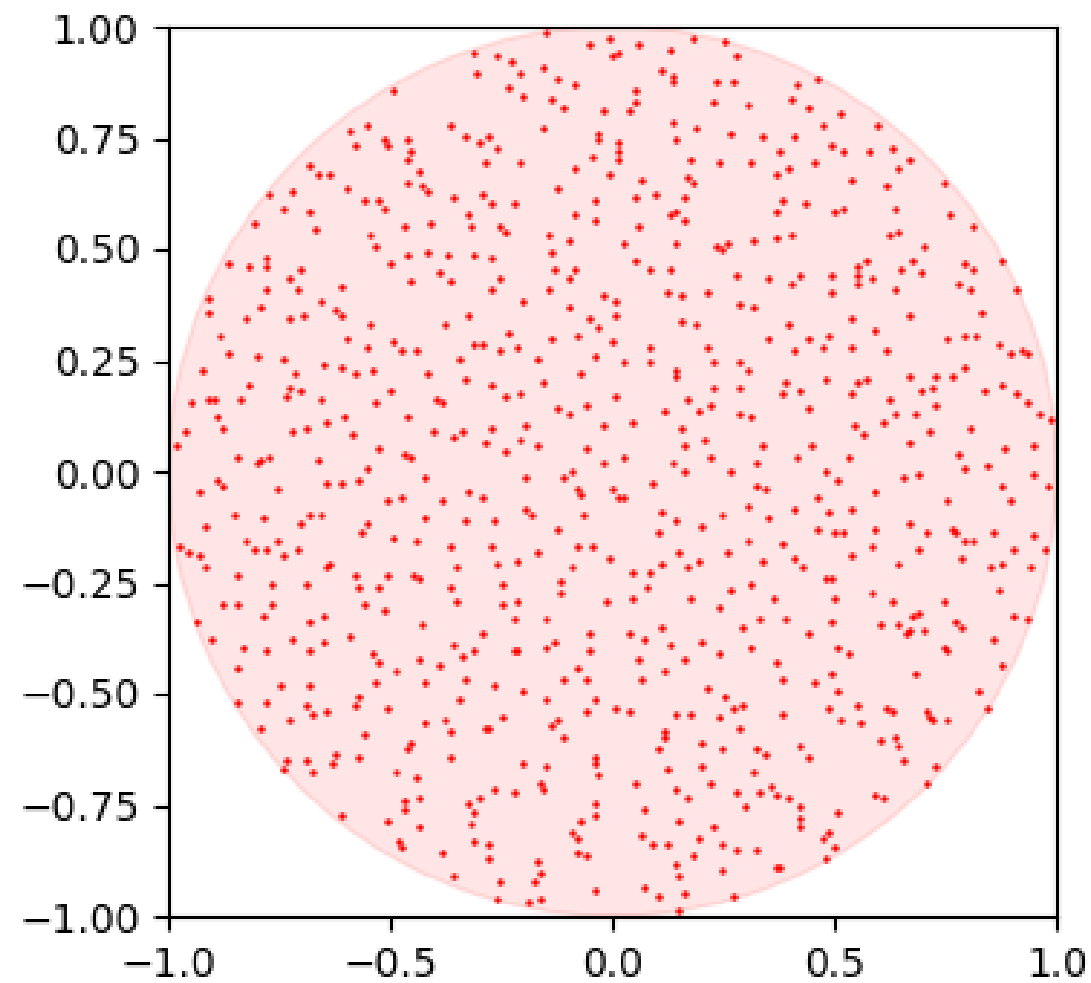
Stratified

Sampling Methods

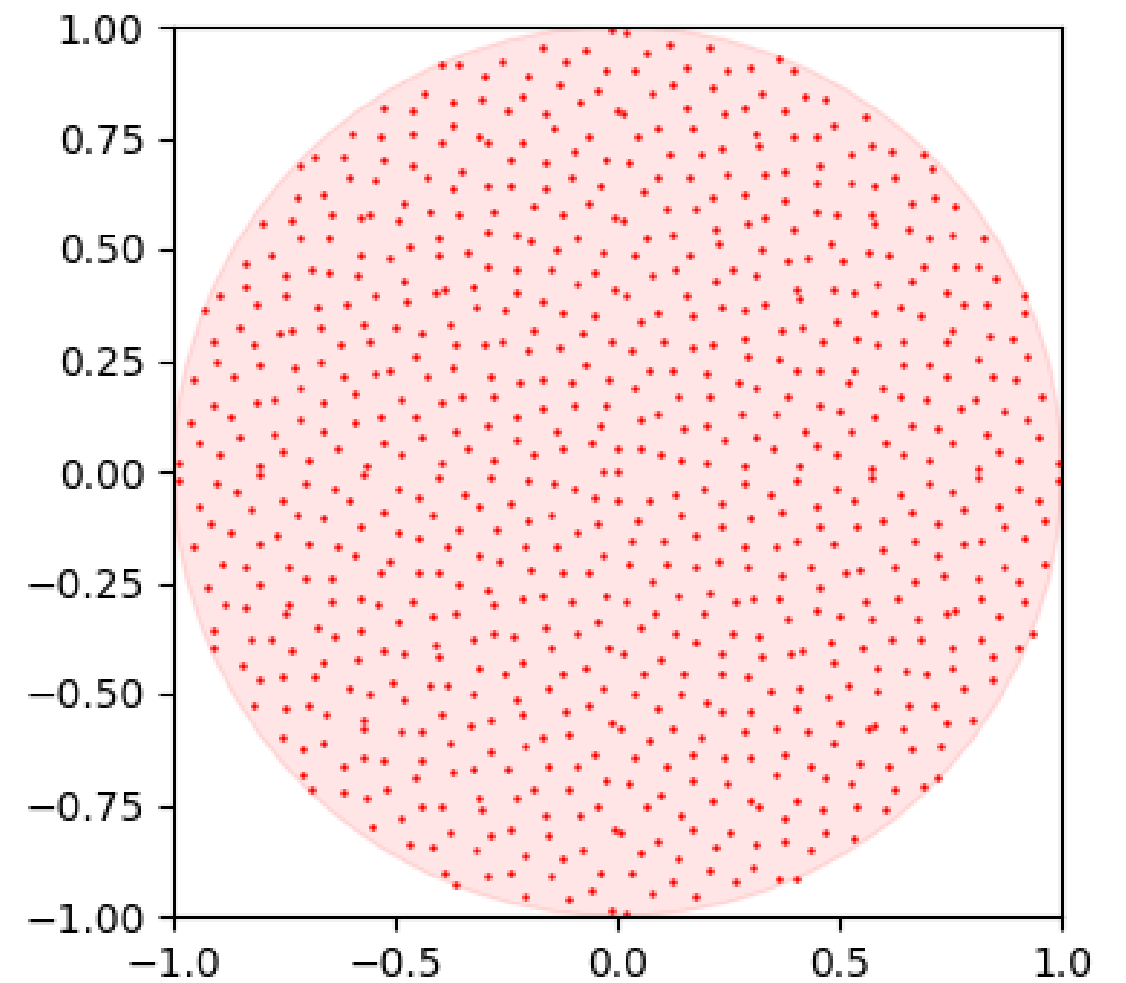
Comparison, N=800



Random



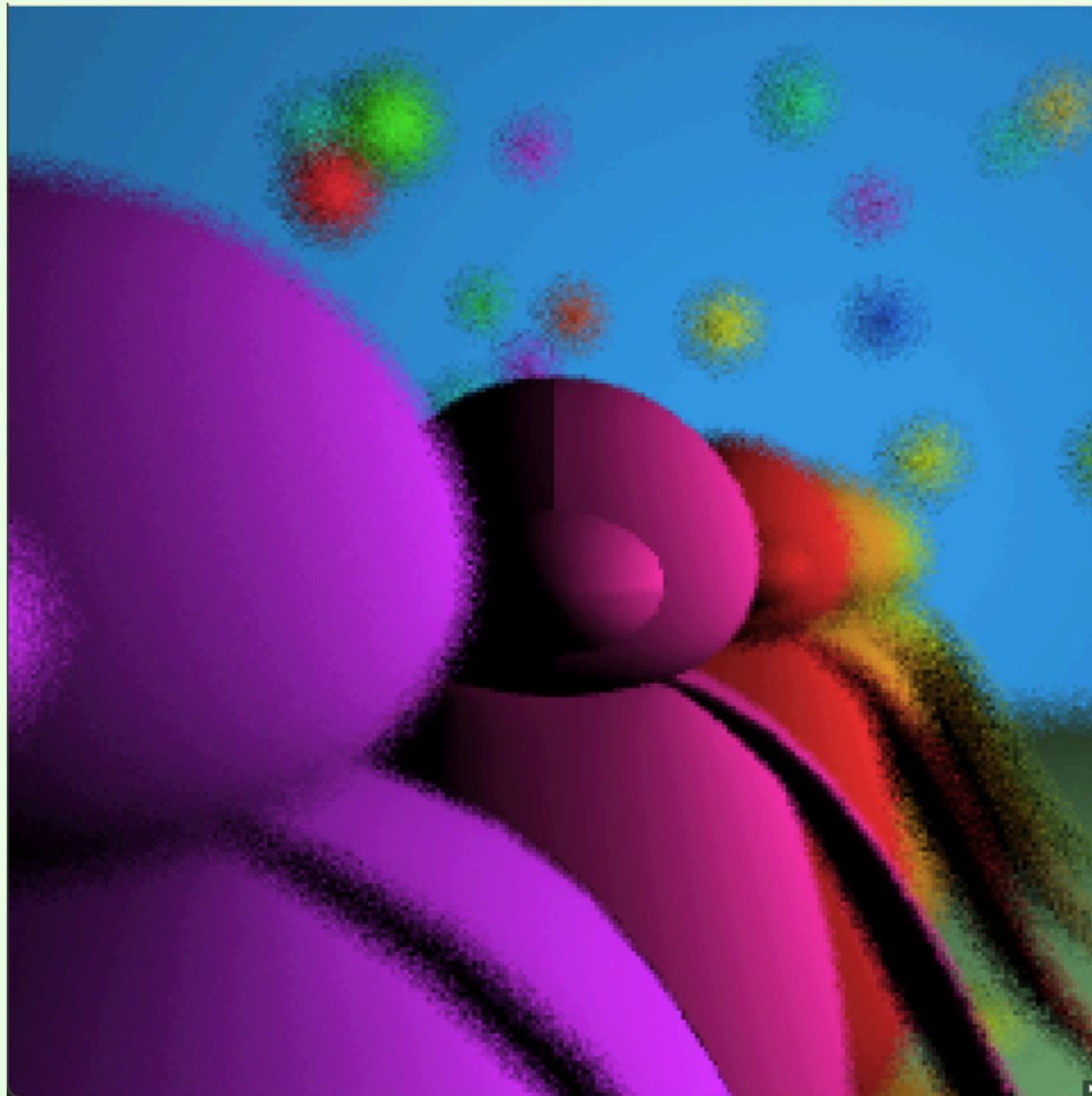
Stratified



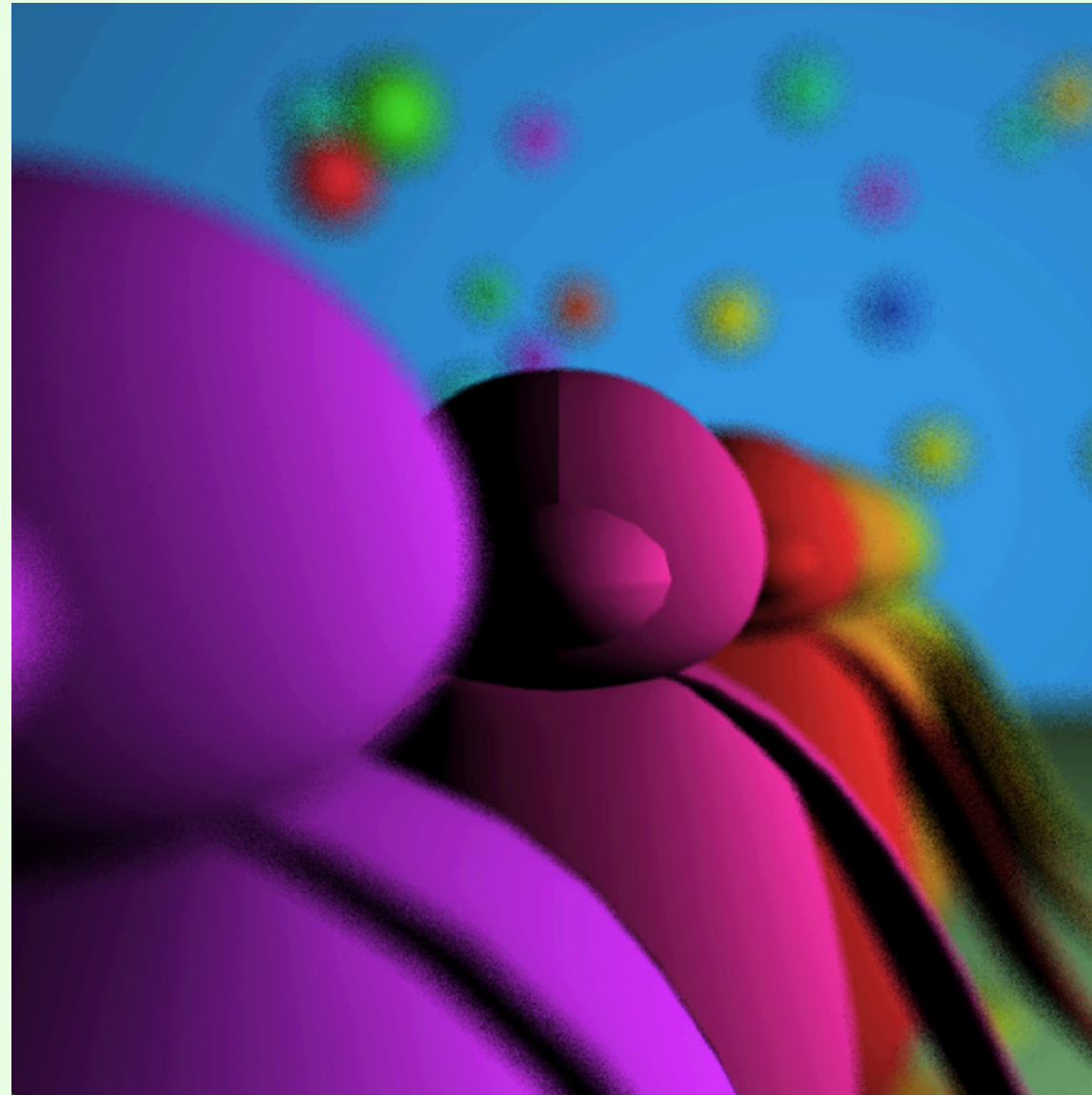
Hammersley

Sampling Methods

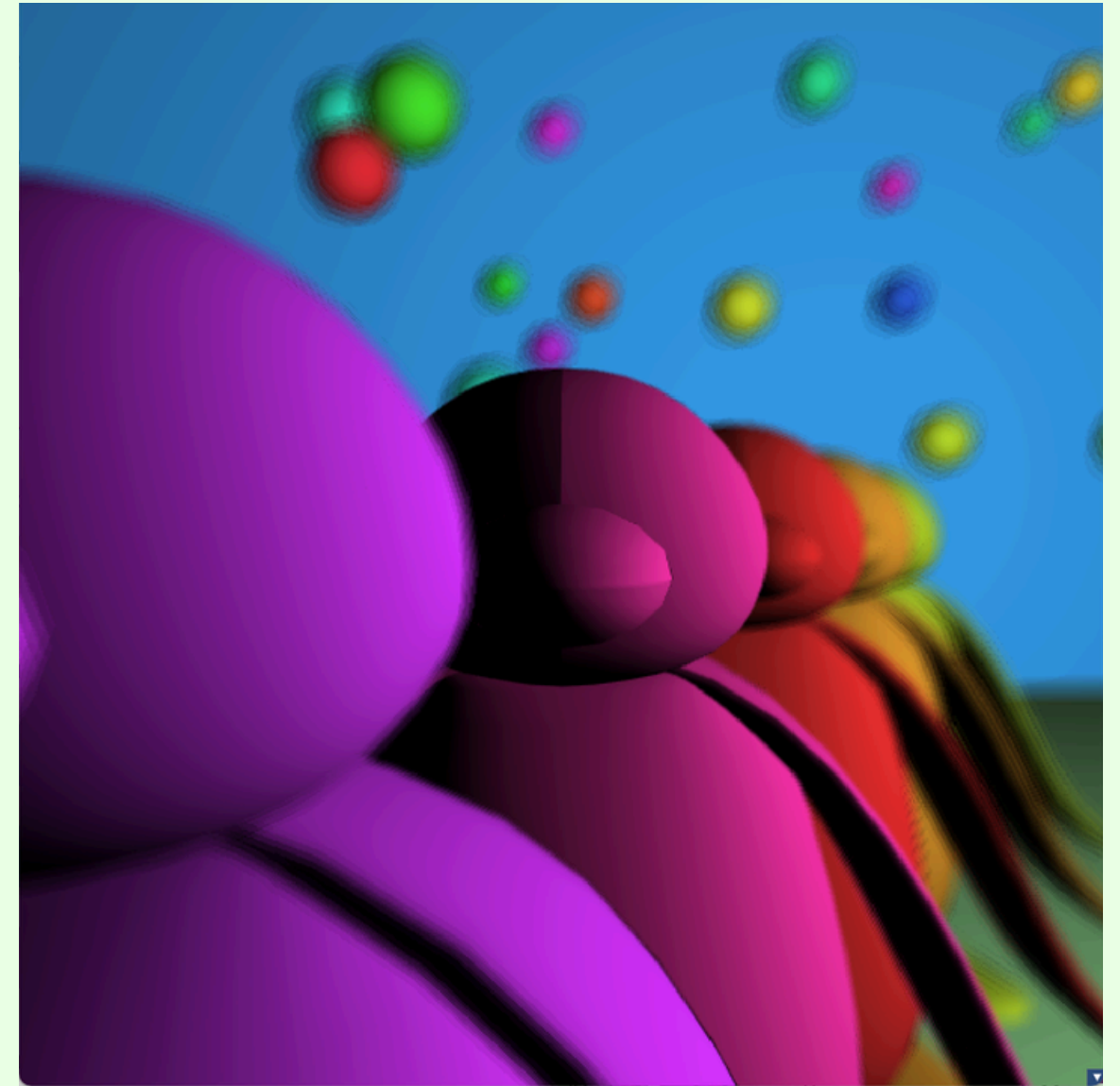
Comparison, N=16



Random



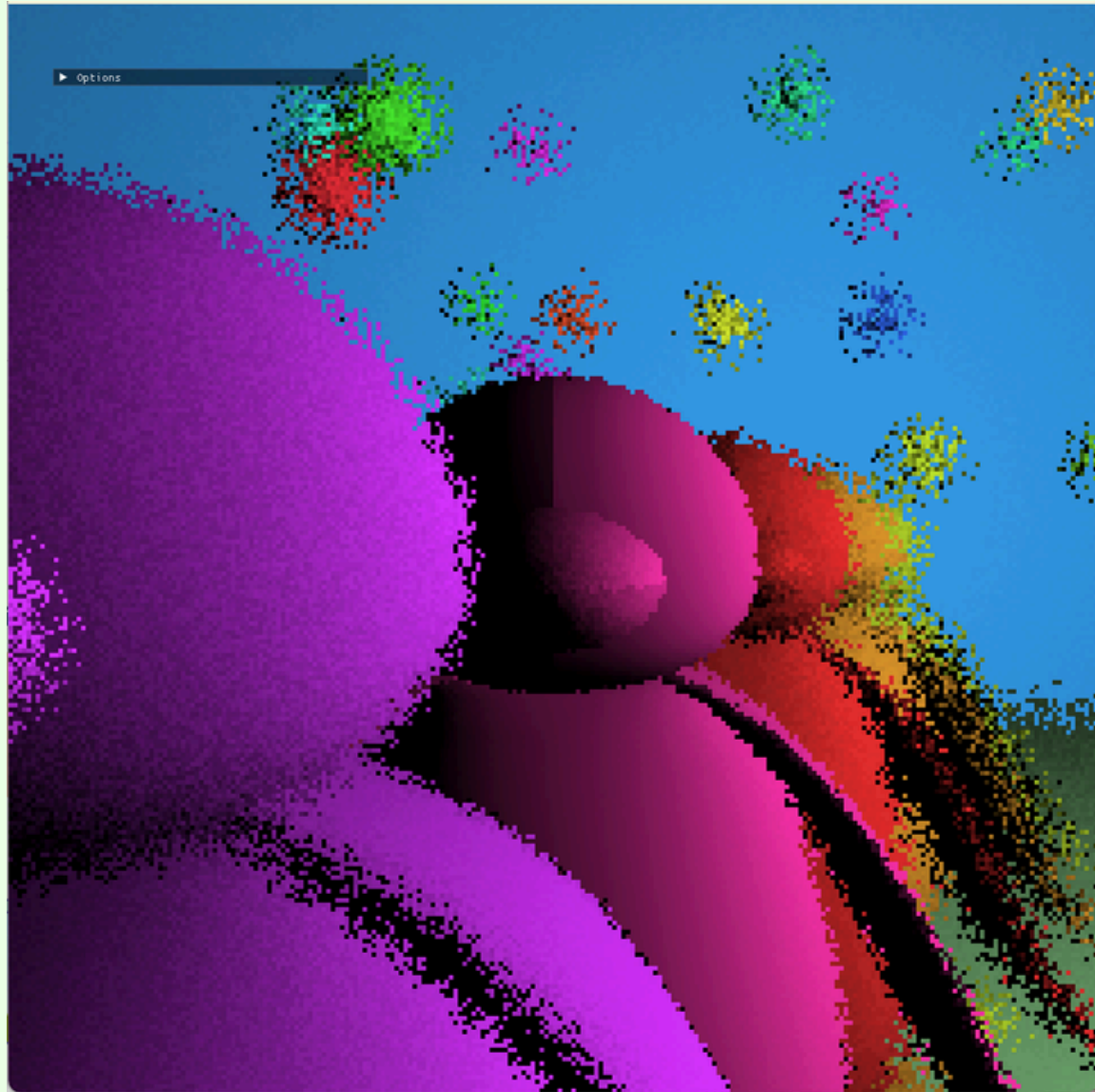
Stratified



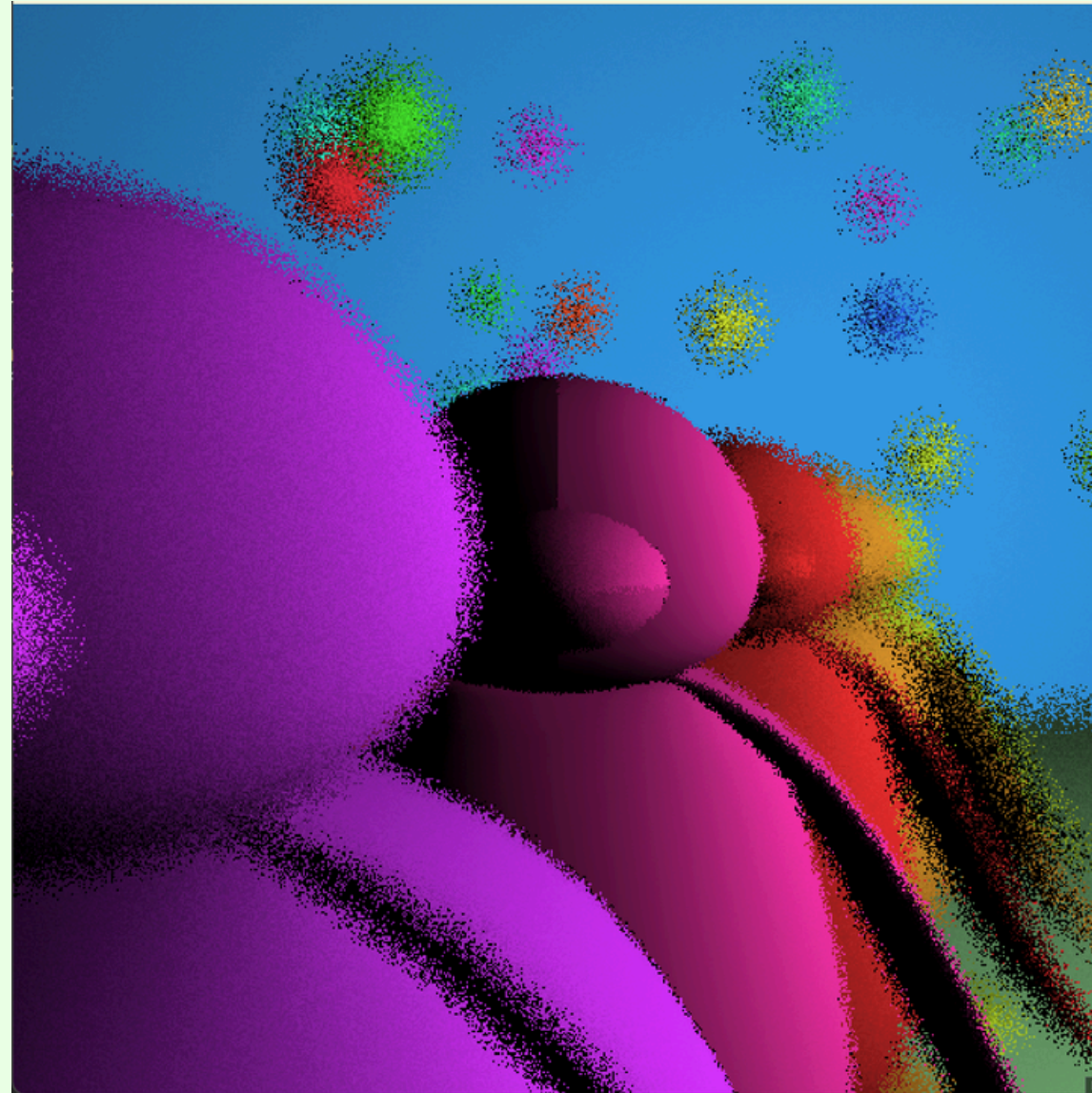
Hammersley

Sampling Methods

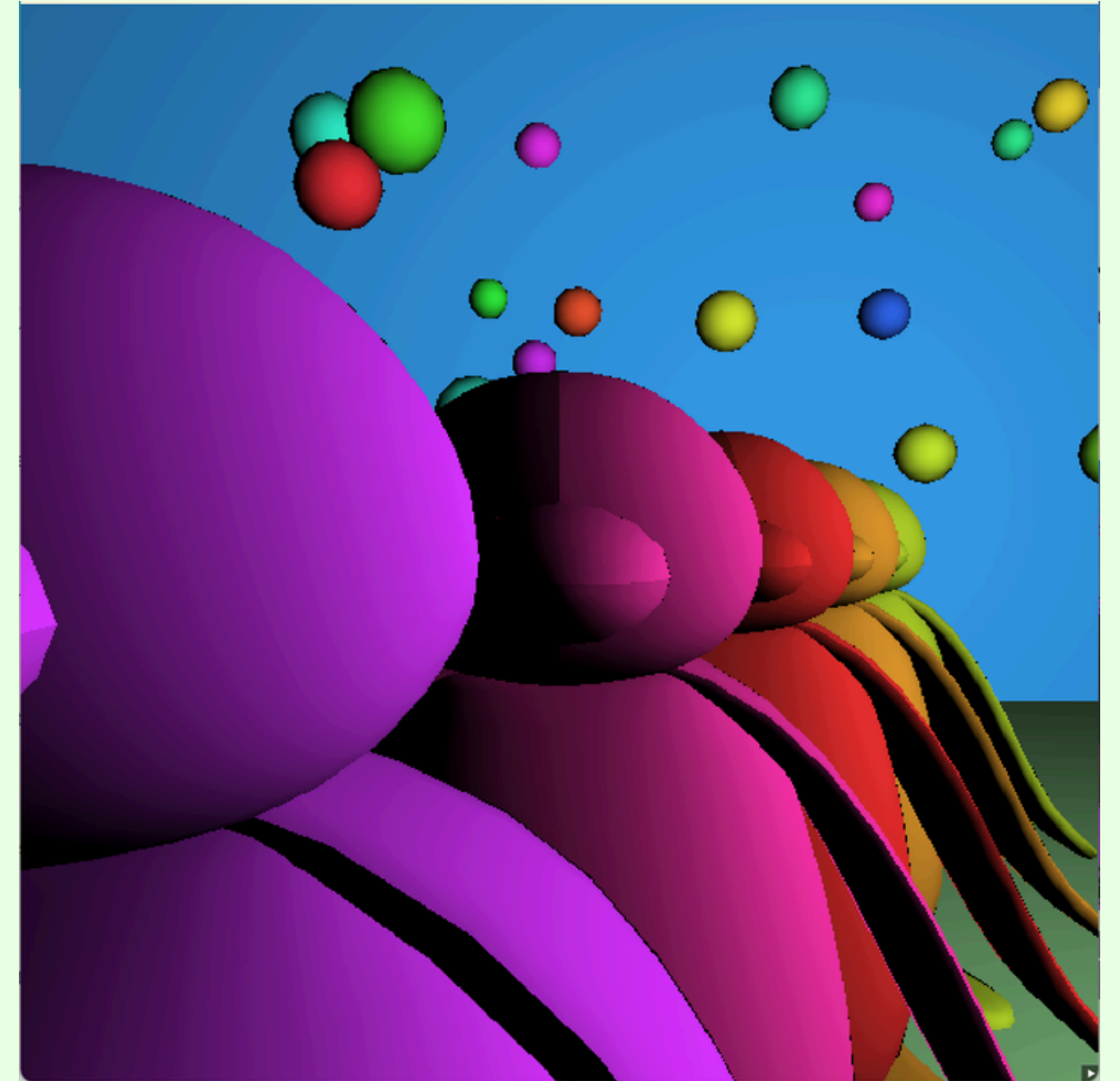
Comparison, $N=1$



Random



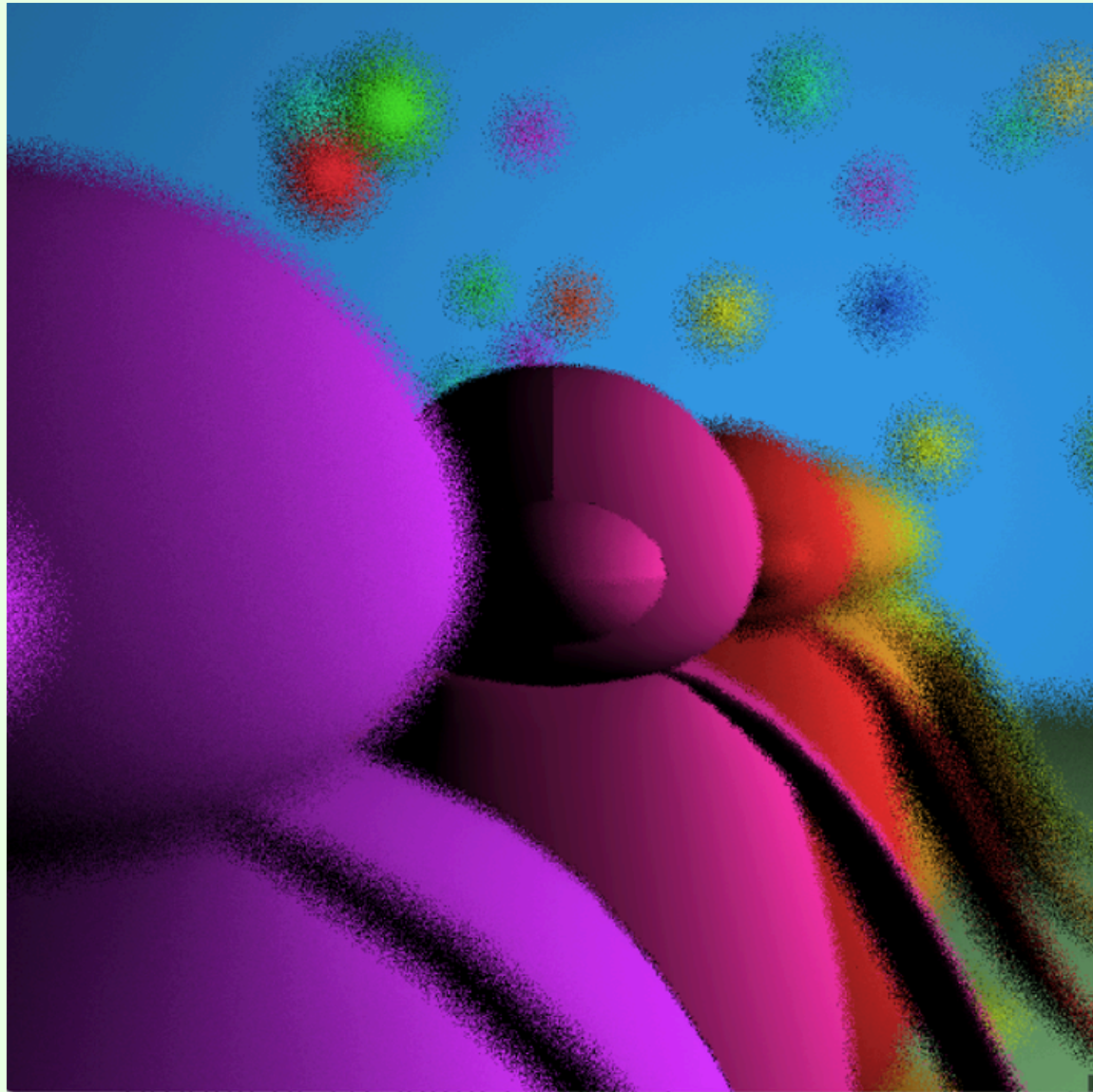
Stratified



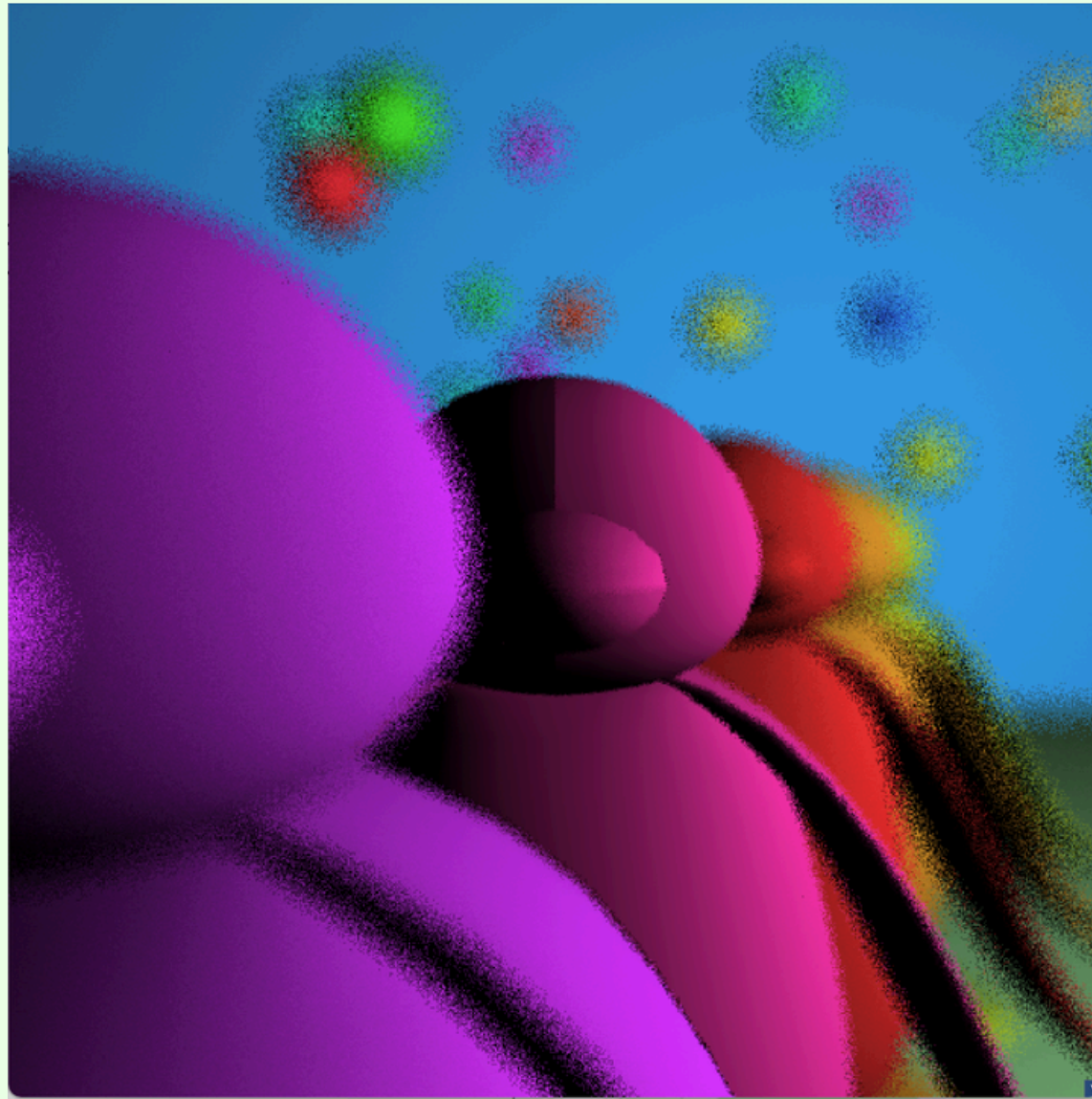
Hammersley

Sampling Methods

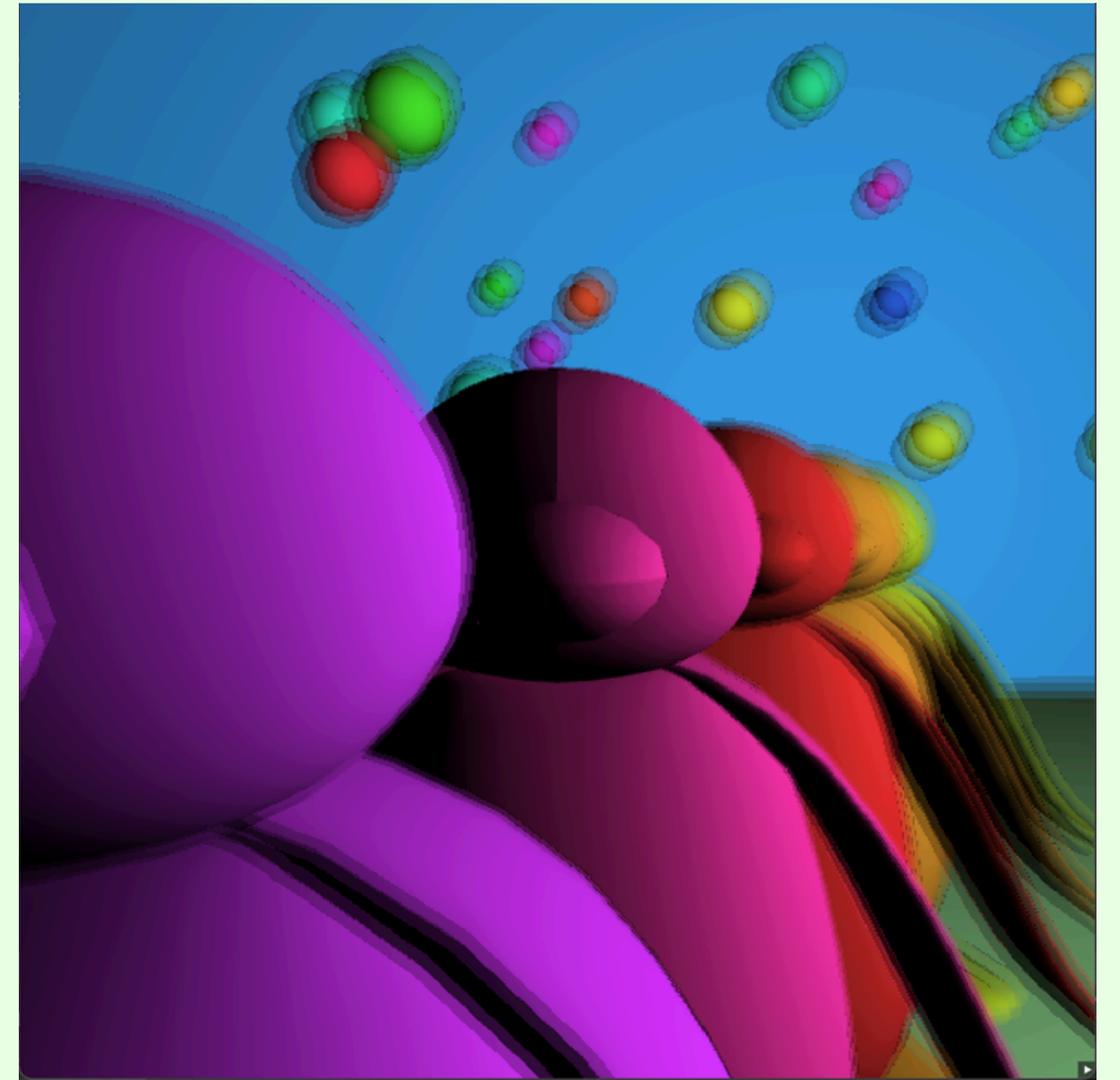
Comparison, $N=4$



Random



Stratified



Hammersley

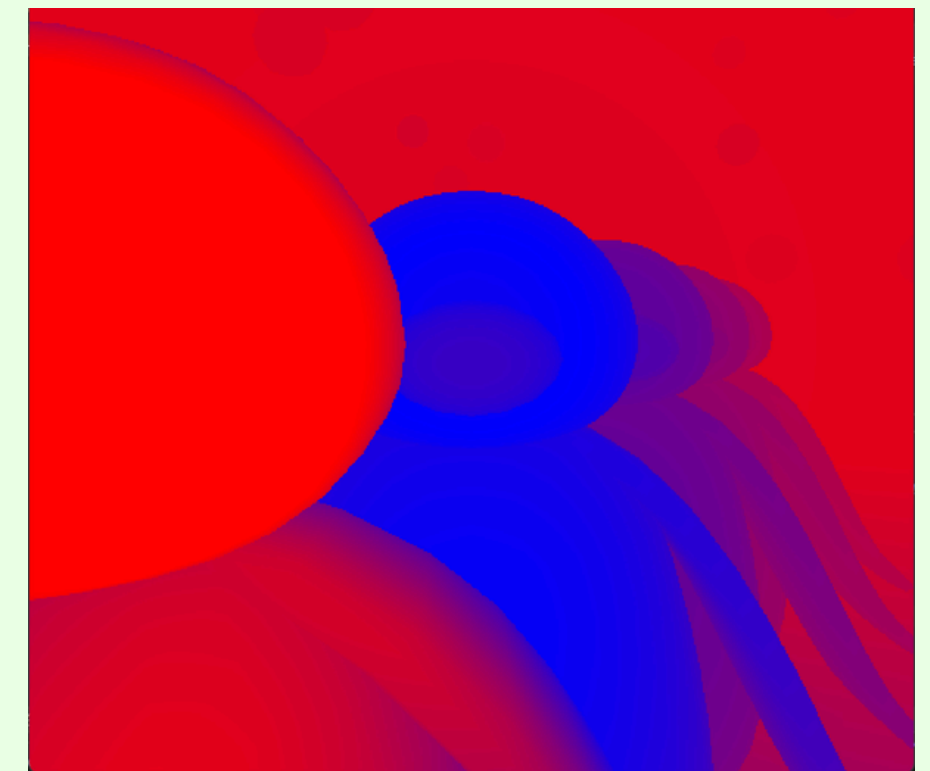
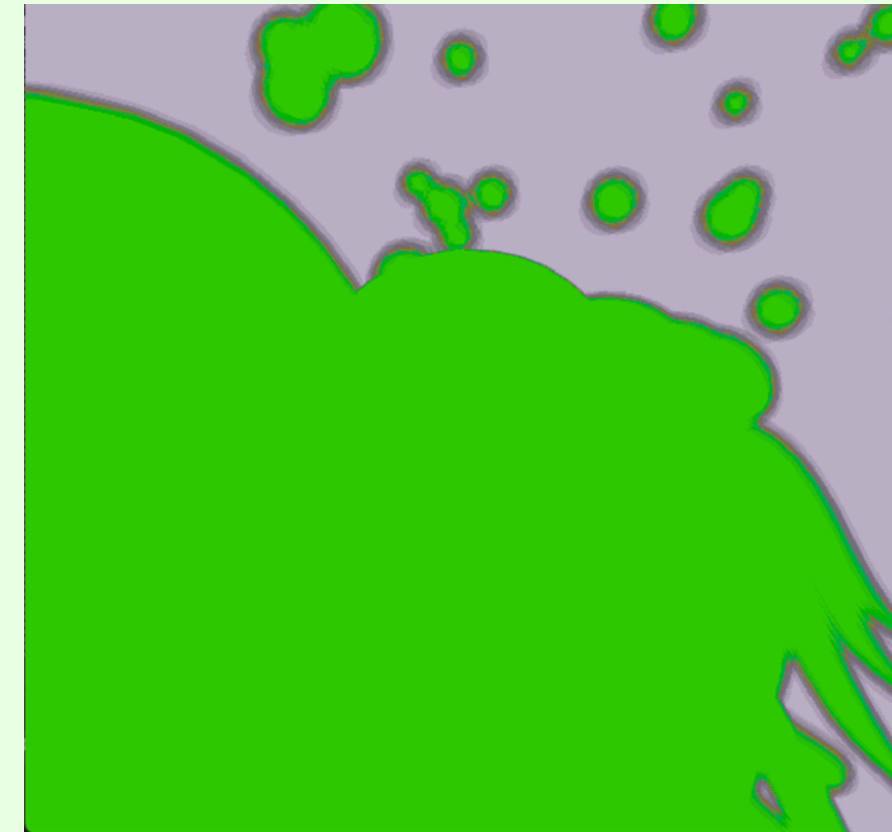
Trying to improve results...

Occlusion by blurry objects in foreground

→ store 2nd hit information

Compute Circle of Confusion

→ if blurry, attenuate 1st layer with
2nd hit



Not effective on its own

criteria too large

→ check neighboring pixels' CoC (heavy)

Up: rays with ≥ 2 intersections

Down: CoC value

Optimization

Get closer to real-time rendering

1

Intersection tests

→ AABB

→ BVH

2

Parallelize DoF
rendering

3

Early exit

Optimization

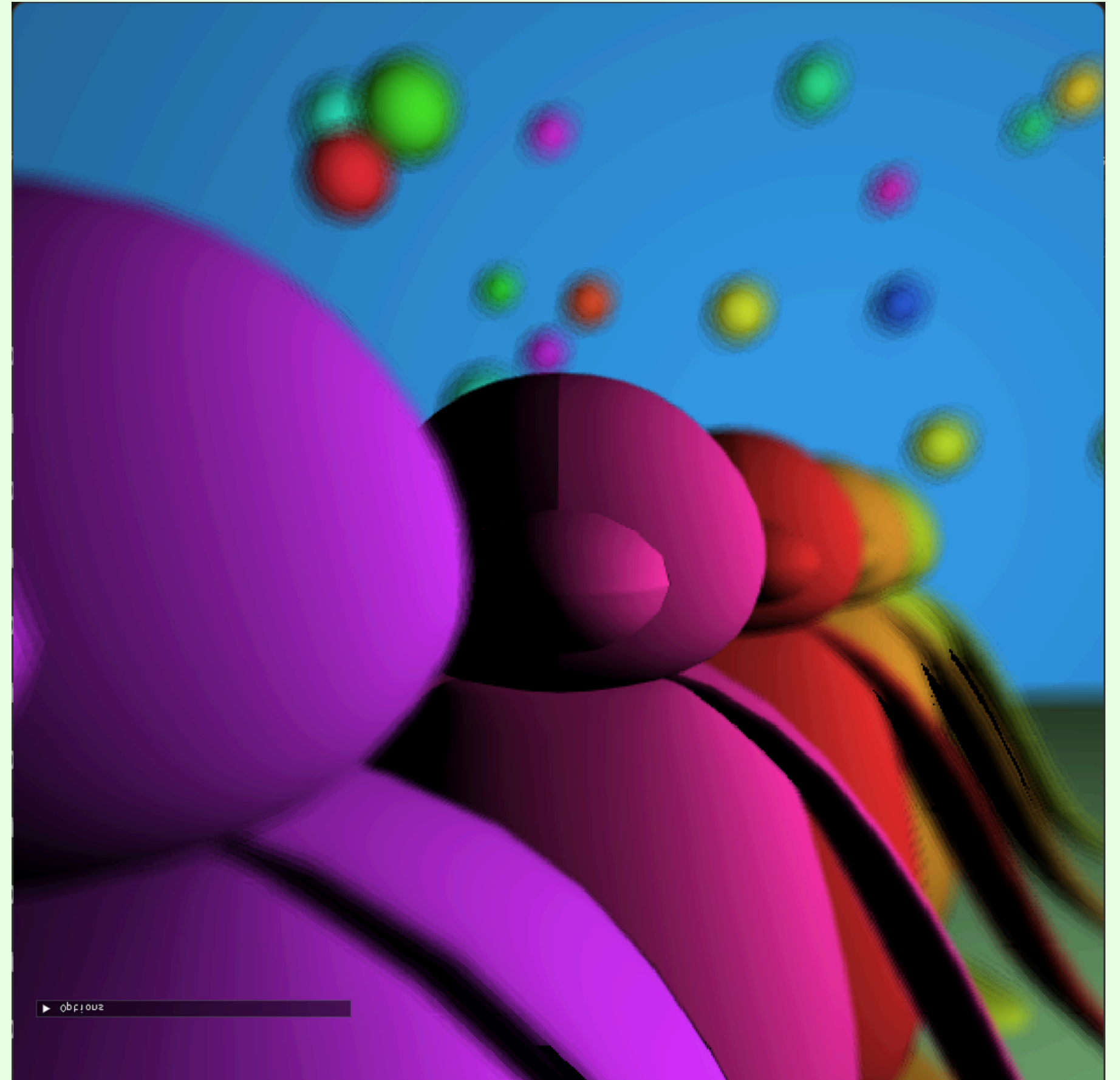
3

Early exit

Stop checking pixel converging
on first sample

→ check luminance variance

$$R * 0.2126 + G * 0.7152 + B * 0.0722 = \text{Luminance}$$



Optimization

3

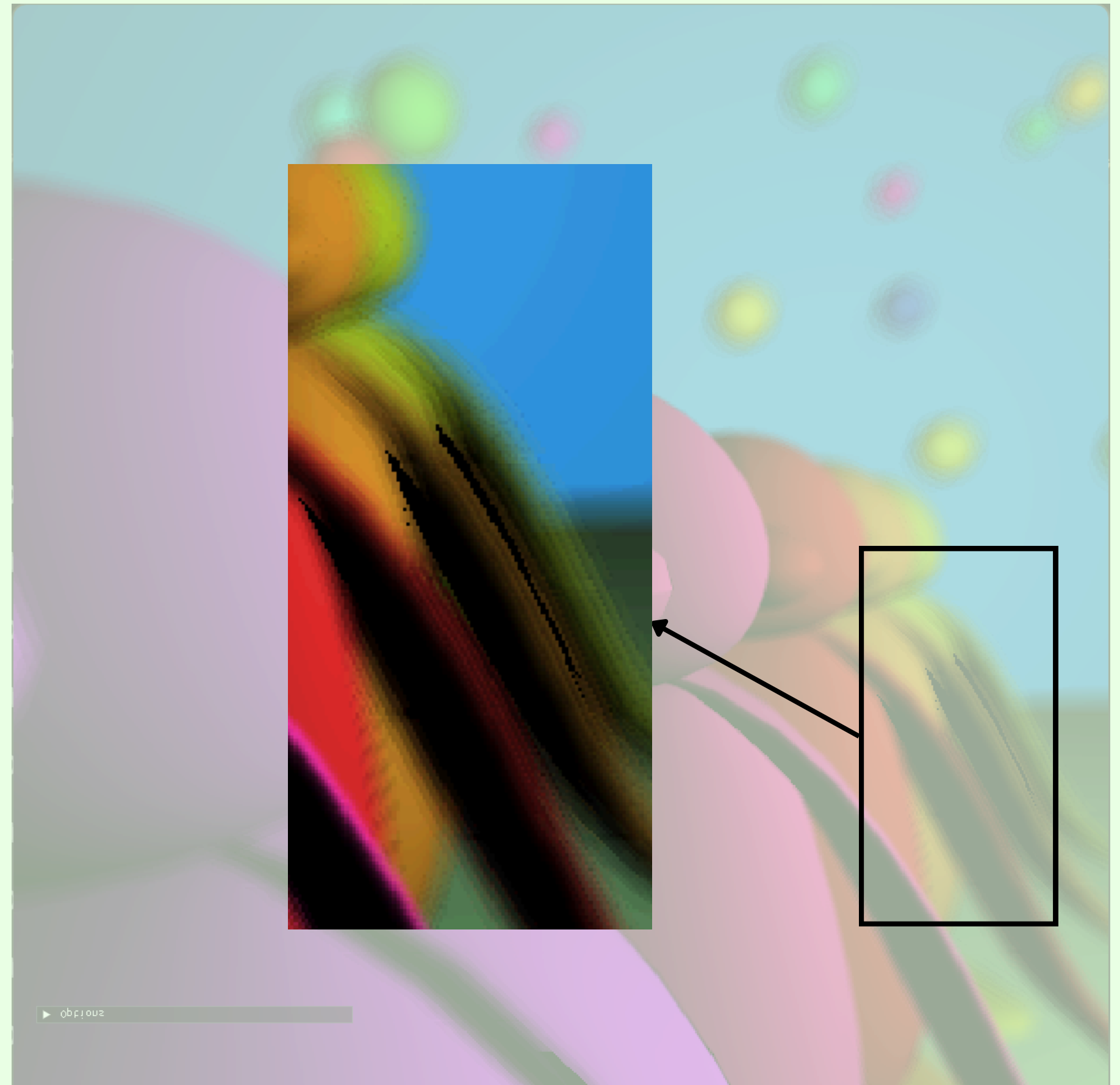
Early exit

Stop checking pixel converging
on first sample

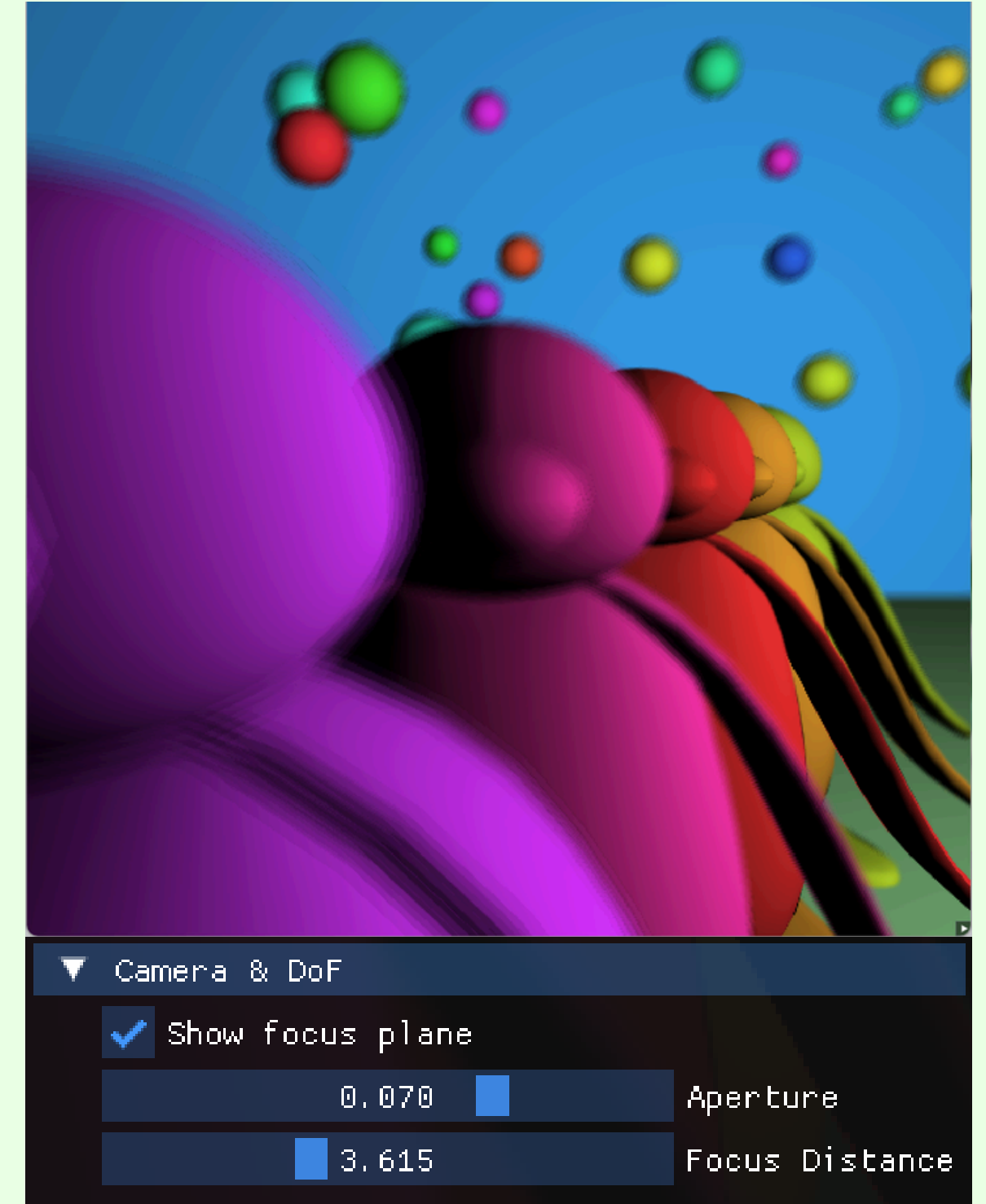
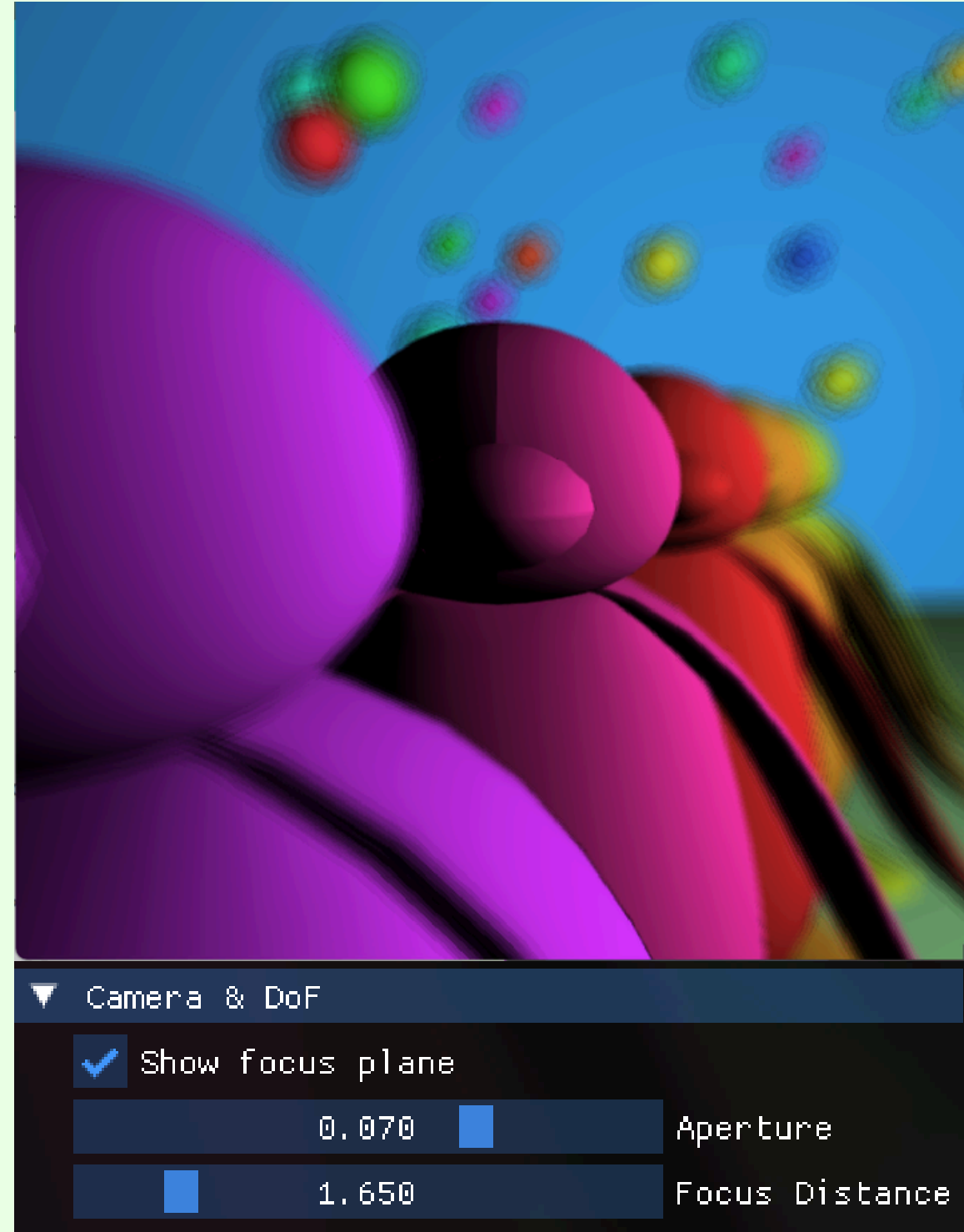
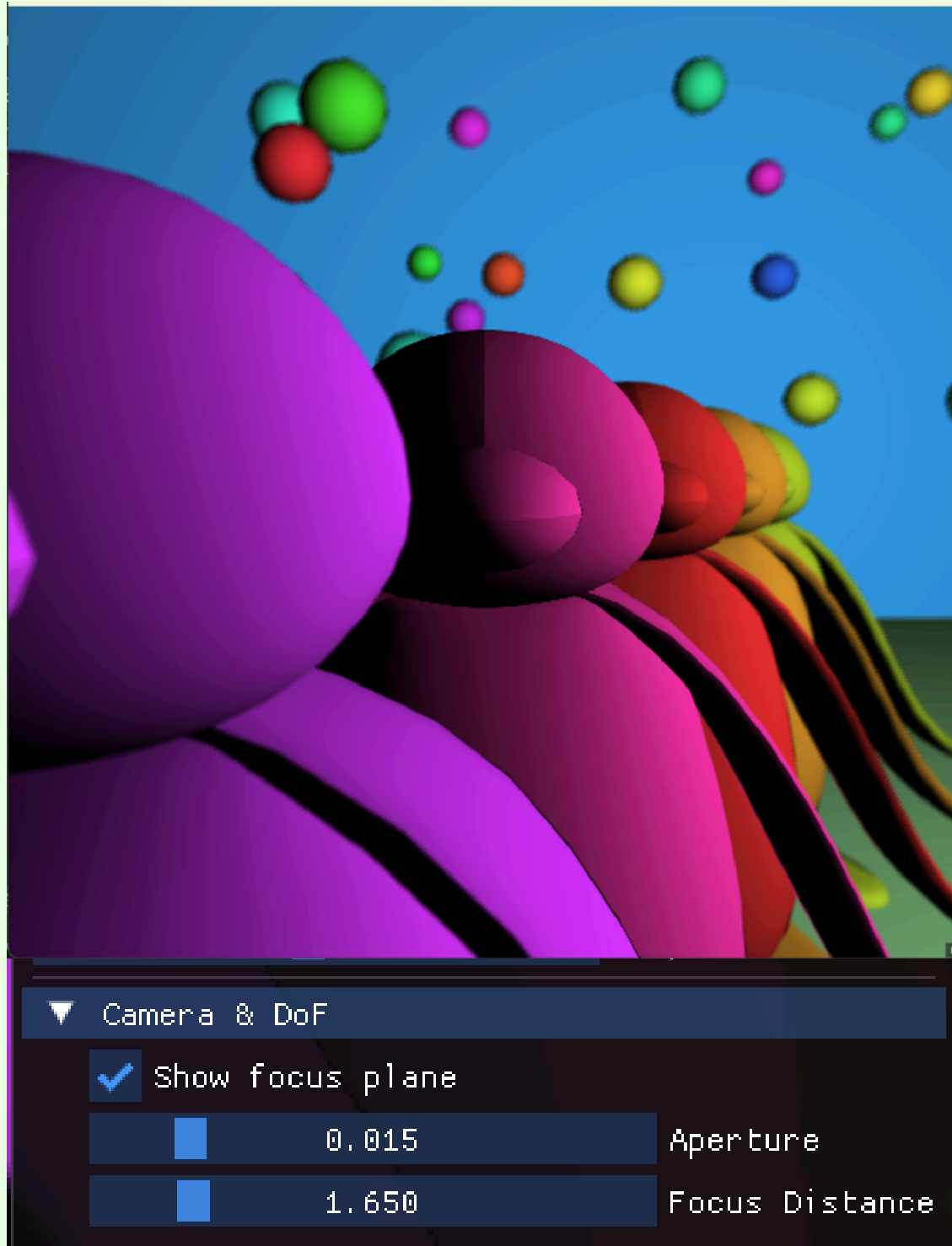
→ check luminance variance

$R * 0.2126 + G * 0.7152 + B * 0.0722 = \text{Luminance}$

Artifacts keep appearing,
tradeoff isn't worth it :/



Results



Further work

Get closer to real-time rendering

1

N-Buffer and scene layering

- major point of original paper
- real-time achievable

2

Handle multiple focus plane

3

Thick lens model

Thank you
for listening :)